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Spatiotemporal Integrated Gradients: Contribution Method for Video Data

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ABSTRACT

Unfinished.

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1. INTRODUCTION

Suppose you have a convolutional neural network (CNN) model trained to predict class labels (digits, people, objects) and that it surpasses humans on such classification tasks. If this model is designed in the fashion of popular networks like ResNet-18, then these models would have hierarchical representations of visual inputs. Thus, simple features such as classical receptive fields would appear in early layers and complex, specific features like faces or a person would appear in later layers. This hierarchical representation of visual stimuli in models is well-founded and closely represents human level organization of feature representation in the ventral visual stream.(Necessary to cite the 30+ papers showing evidencing this)

The aforementioned logic assumes that the CNN has not learned a trivial look-up table for the inputs and their labels, which would make the following research questions and line of reasoning pointless. What representations within the CNN contribute to correct identification? What internal layer functions necessitate successful object categorization? And most importantly: what does each layer of the model see when given an input? Such line of questioning are at the core of explainable AI (XAI). XAI is closely related to the field of mechanistic interpretability, putting the 'science' in computer science and elucidating principles and laws of neural network computations. In simple terms, XAI and MechInterp have a shared goal of parsing how neural network models perform well-defined tasks.

1.1. *Background on Interpretability Methods*

For image classification tasks, interpretability tools like Gradient-weighted Class Activation Mapping (GradCAM) provides visual explanations for classification tasks. This method works by computing the gradients from an input with respect to the activations of the final convolutional layer, indicating the importance of each feature map. These gradients are averaged, summed, then thresholded to combine these features into a single map of important regions overlayed onto the original input. This method gives a visual representation of the input features that are most salient to the network. The XAI community has expanded on this method of interpretability in vision networks with GradCAM++, ScoreCAM, and Integrated Gradients. The last of which I will expand upon for the rest of this document.

Integrated gradients (IGs), is a gradient based method that calculates a path integral from baseline to stimulus to any layer of interest. This is accomplished through making a path, collecting gradients, approximating the integral from baseline to the stimulus. This explains which part of the network contributes to the output. Further, using integrated gradients enables network contribution mapping: identifying channel specific attributions for convolutional layers. By leveraging the fundamental theorem of calculus: the integral and differentiation are inverse operations; integrated gradients asserts linearity, sensitivity, and completeness. Thus, integrating gradients through a neural network is a more rigorous attribution method for explaining model computation and functions. The original method was shown to explain model choices in image, language, and molecular data–lending to it's broad applicability to explaining how networks arrive at their output.

1.2. *Motivation for Sequential Integrated Gradients*

We will expand the integrated gradient method in two ways: using (1) models trained to predict neuronal firing rates in response to (2) video inputs that the neurons observed. The following research questions can be asked: What does

each layer of the neural network observe as determined by contribution maps from top contributing channels? What cell types are necessary for emergent cell type phenomenology like direction selectivity, omitted stimulus response, or object motion detection?

The Integrated gradients method—while powerful, has been shown to work best on static datasets. This method accumulates gradients from baseline to the input, which allows the practitioner to glean what features of the input contribute most to model output. The question lies in what a good baseline should be for calculating this integral. In one case integrating from black, or zero, would be analogous to seeing what parameters of the model are salient as the full image comes into view. It can be argued that this approach biases the explanation, thus, averaging the path across many baselines (e.g. variable starting points between black and gray) up to the full image gives a better estimate of the path attribution.

Applying this method to videos would require a per-frame IG computation or averaging across batches of frames—which would result in computationally intensive and low quality estimations of the attributions. To sequentially compute integrated gradients accurately, we must compute a spatiotemporal integral. Sequences of frames must be processed in mini-batches, permitting both spatial and temporal investigation of network outputs to video data.

Given that spike trains and video inputs are time series data, it is important to know the trajectory of the contributions over sequences of events, allowing for better quality attributions in these data domains. Ideally, we want to maintain the original benefits of the standard integrated gradient application: completeness, sensitivity, and linearity. For example, if features from a sequence of frames have the same impact as the baseline input, then the resultant attribution value should be zero. However, adapting the original method to spatiotemporal data shows whether those features were truly ineffective at perturbing the model or if took a path that led to null result. The standard IG method does not allow us to observe the positive and negative cancellations that occur as a result of competing effects on model outputs. Additionally, accounting for the time allows us to elucidate time dependencies inherent to video and spike train data, preventing the independence assumption that state of the art methods use.

2. RELATED WORK

2.1. *Explainable Artificial Intelligence for Time Series Data*

In time-series data, the relationships between data points are influenced by their order and historical context. Naively applying the standard integrated gradient method and treating observations as independent, often fail to capture these dynamics, leading to the creation of Out-of-Distribution (OOD) intermediate inputs. Recent methods address this by incorporating temporal dependencies, primarily focusing on local feature masking or perturbation for final prediction attribution The following methods based on Integrated Gradients (IG) offer complementary approaches for using neural network models on time-dependent data. Recent efforts have applied IG to various sequential and multi-dimensional domains:

- **Temporality-Aware Integrated Gradients (1D): TIMING** (Jang et al., 2025) explicitly addresses the OOD problem in $1D$ time series data, demonstrating that while conventional IG is theoretically sound, its naive application ignores temporal relationships. **TIMING** enhances IG by incorporating a temporal-aware path to improve directional attribution on data like medical records. In this extension they take subsamples of the time sequence data with which to compute the Riemann sum over a sufficient number of steps, then sum the attributions. While this method allows for higher quality temporal estimates of attribution, this method loses completeness as the section they sampled are not acquired from a linear path, but a randomly selected sequence of the history. This method addresses the OOD issue in the sacrifice of completeness.

- **Integrated Gradients for Spatio-Temporal Data (3D):** Work on precipitation nowcasting (Ahangama et al., 2025) demonstrates the feasibility of applying standard **Integrated Gradients** to high-dimensional spatio-temporal image sequences (e.g., satellite video data) for region-of-interest identification in weather data to predict the next half hour of weather conditions. However, this method relies on the *simultaneous straight-line interpolation* path, which does not address the OOD problem inherent in video. Instead, they essentially apply the integrated gradient method over every Nth frame, accounting for time, but possibly ignoring the trajectory that the data took. This paper highlights that there may be an acceptable limit of estimation that does not require practitioners to compute gradients for the entire sequence of input frames.

- **Sequential Integrated Gradients (SIG) in NLP:** The work by Enguehard (2023) introduces **Sequential Integrated Gradients (SIG)** for language models, calculating token importance by interpolating only the token of interest while *fixing the context* of all other words. This confirms the conceptual necessity of a sequential, context-preserving path for attribution in discrete, contextual data.

The existing use of the integrated gradient method reveals a critical distinction in objectives: most methods (masking, perturbation) focus on attributing features to the **final prediction score**. Our approach, the **Piecewise Sequential Integrated Gradients (SIG)**, is novel because we combine the necessity of a causally-constrained path with attribution to **intermediate channel units** in a biologically-constrained model. This addresses two gaps simultaneously:

Methodological Gap: We overcome the OOD and causal inconsistency of naive IG on video by defining a **piecewise linear path** that fixes the history frames $\mathbf{x}_1$ to $\mathbf{x}_{t-1}$ when integrating frame $\mathbf{x}_t$. This ensures the attribution of each time step is measured within a temporally coherent context.

Scientific Gap: The core objective is not merely prediction attribution, but to leverage the model as a platform for neuroscientific inquiry. By performing **Output-to-Hidden** attribution, we investigate whether the learned internal units of the model trained to predict V1 spike trains have **temporal correspondence** to known biological phenomenology.

This framework is essential for probing functions that are inherently time-dependent in the visual system. This functional identification is critical because many foundational operations in the visual system are inherently **time-dependent** and thought to arise from the interaction of distinct cell types. For example, phenomena like **polarity reversal**, **latency encoding**, **frequency doubling**, **motion anticipation**, **object motion detection**, and **omitted stimulus response** emerge from the precise spatiotemporal summation and interaction of inputs from various upstream pathways. Our approach provides a tool to extract the temporal contribution map from many cell types simultaneously, which is essential to explain the emergence of these complex, predictive, and time-dependent phenomena.

3. METHODS

3.1. *Standard Integrated Gradients*

The standard integrated gradients method computes attributions for static images by integrating gradients along a path from a baseline to the input. The attribution for feature i is given by:

$$\mathrm{IG}_i(x) = (x_i - x_i') \times \int_{\alpha=0}^{1} \frac{\partial F(x' + \alpha \times (x - x'))}{\partial x_i} d\alpha \tag{1}$$

where F is the model, x is the input image, x' is the baseline (typically a black image), and α parameterizes the path from baseline to input

Algorithm 1 Standard Integrated Gradients (IG) Method

Require: F: model; x: input image; c: target neuron/class; x': baseline (e.g., black image); n: number of interpolation steps
Ensure: A: attribution map showing spatial importance

```
Initialize A ← 0 with same shape as x
Generate interpolation coefficients {α_1, α_2, ..., α_n} evenly spaced in [0, 1]
for each α in {α_1, ..., α_n} do
    Compute interpolated input: x_α ← x' + α × (x − x')
    Enable gradient tracking for x_α
    Forward pass: y ← F(x_α)
    Select target output: y_c ← y[c]
    Backward pass to obtain gradients: g_α ← ∂y_c / ∂x_α
    Store g_α
end for
Approximate path integral using trapezoidal rule:
    IG ← 0
for i = 1 to n − 1 do
    IG ← IG + 0.5 × (g_i + g_{i+1})
end for
Compute mean integrated gradient: IG_mean ← IG/n
Compute final attribution: A ← (x − x') ⊙ IG_mean
return A
```

3.2. *Sequential Integrated Gradients*

For video sequences, we introduce Sequential Integrated Gradients (SIG), which extends the standard IG method to handle temporal data. The key insight is to compute a piecewise path where each frame is revealed sequentially. For a video with T frames, the attribution for feature i in frame t is:

$$\mathrm{SIG}_{i(t)}(x) = \sum_{k=1}^{T} \mathrm{IG}_{i(t),k}(x) \tag{2}$$

$$\mathrm{SIG}_{i(t),t} = (x_{i(t)} - x_{i(t-1)}) \times \int_{(t-1)/T}^{t/T} \frac{\partial F(r^*(\alpha))}{\partial x_{i(t)}} \cdot T \, d\alpha \tag{3}$$

3.2.1. *Defining the Piecewise Path*

We define the path $r^*(\alpha)$ for $\alpha \in [0, 1]$, where the total path is divided into T equal-length segments, each of length $1/T$.

Let $\mathbf{X}_t^{\text{partial}}$ be a "partial input" where the first t frames are set to their true values $(x_1, \ldots, x_t)$ and the remaining $T - t$ frames are set to their baseline values $(x'_{t+1}, \ldots, x'_T)$. We define $\mathbf{X}_0^{\text{partial}} = \mathbf{X}'$.

For the t-th segment, corresponding to $\alpha \in \left[\frac{t-1}{T}, \frac{t}{T}\right]$, we transition from the partially revealed input $\mathbf{X}_{t-1}^{\text{partial}}$ to the next partially revealed input $\mathbf{X}_t^{\text{partial}}$.

The scaled parameter β within the segment $[0, 1]$ is defined as:

$$\beta = \alpha \cdot T - (t - 1), \quad \text{for } \alpha \in \left[\frac{t-1}{T}, \frac{t}{T}\right] \tag{4}$$

The path $r^*(\alpha)$ is then defined by:

$$r^*(\alpha) = \mathbf{X}_{t-1}^{\text{partial}} + \beta \cdot (\mathbf{X}_t^{\text{partial}} - \mathbf{X}_{t-1}^{\text{partial}}) \tag{5}$$

Algorithm 2 Spatiotemporal Integrated Gradients (SIG) Method

Require: F: model; X: input video with T frames; c: target neuron/class; X': baseline video (all black frames); s: number of interpolation steps per segment

Ensure: Spatiotemporal attribution tensor $\mathrm{SIG}_{i(t)}$ for each feature i in frame t

Initialize $X' \leftarrow$ video of T black (zero) frames
Define partial input states: $X_0^{\text{partial}} \leftarrow X'$, $X_T^{\text{partial}} \leftarrow X$
Initialize storage for frame-wise attributions $\mathrm{SIG}_{i(t)}$
for $t = 1$ to T **do**
  Define segment boundaries on the integration path:
  $\alpha \in [\frac{t-1}{T}, \frac{t}{T}]$
  Frames $1, \ldots, t-1$ fixed at true values, frame t transitions from baseline to true value, frames $t+1, \ldots, T$ remain baseline.
  Generate s interpolation steps $\{\beta_1, \beta_2, \ldots, \beta_s\}$ where $\beta_j = \frac{j}{s}$
  for $j = 1$ to s **do**
    Construct partial video $X^{\text{partial}}(j)$:
    For frames $1, \ldots, t-1$: $x_k^{(j)} = x_k$
    For frame t: $x_t^{(j)} = x'_t + \beta_j \times (x_t - x'_t)$
    For frames $t+1, \ldots, T$: $x_k^{(j)} = x'_k$
    Stack all frames as required by F (e.g., concatenate along channel dimension)
    Forward pass: $y \leftarrow F(X^{\text{partial}}(j))$
    Select target output: $y_c \leftarrow y[c]$
    Backward pass to obtain gradients: $g_t^{(j)} \leftarrow \frac{\partial y_c}{\partial x_t^{(j)}}$
    Store $g_t^{(j)}$
  end for
  Approximate integral for frame t using trapezoidal rule:
  $\mathrm{IG}_t \leftarrow 0$
  for $j = 1$ to $s-1$ **do**
    $\mathrm{IG}_t \leftarrow \mathrm{IG}_t + 0.5 \times (g_t^{(j)} + g_t^{(j+1)})$
  end for
  Scale by segment length: $\mathrm{IG}_t \leftarrow T \times \frac{\mathrm{IG}_t}{s}$
  Compute frame attribution:
  $\mathrm{SIG}_{i(t)} \leftarrow (x_{i(t)} - x'_{i(t)}) \odot \mathrm{IG}_t$
  Store $\mathrm{SIG}_{i(t)}$ in attribution tensor
end for
return Complete attribution tensor SIG containing spatiotemporal importance scores satisfying $F(X) - F(X') = \sum_{t,i} \mathrm{SIG}_{i(t)}$

3.3. *Contribution Map Visualization*

After computing attributions using either IG or SIG, we can identify which channels in a given layer contribute most to the model's predictions. This enables visualization of what the model "sees" at different layers.

4. RESULTS

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5. DISCUSSION

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6. CONCLUSION

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Algorithm 3 Contribution Map Generation

Require: Attribution tensor A (from IG or SIG), target layer L, number of top channels K, original input dimensions
Ensure: Visualization maps showing top contributing channels

```
attributions_L ← Extract attributions for layer L from A
if using SIG then
    ▷ Attributions have temporal dimension
else
    ▷ Attributions are static
end if
contributions ← []
for each channel k in layer L do
    score_k ← Σ_{i,j} |attributions_L[k, i, j]|    ▷ Sum over spatial dims
    Append score_k to contributions
end for
Sort contributions in descending order
top_channels ← Select top K channels
threshold ← mean(contributions) + std(contributions)
Filter channels: keep only those with score_k > threshold
for each selected channel k do
    activation_k ← Extract activation pattern from layer L during forward pass
    activation_resized_k ← Resize to input dimensions (bilinear interpolation)
    Normalize activation_resized_k to [0, 1] range
end for
Create tiled visualization grid of top channel activations
if temporal data then
    Create separate visualizations per time point or average across time window
end if
return Visualization maps
```

AUTHOR CONTRIBUTIONS

JCW

Facilities:

Software:

APPENDIX

A. FUNDING

Empty

B. AUTHORS' CONTRIBUTIONS

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C. CONFLICT OF INTEREST

The author declares no conflict of interest.

D. DATA AVAILABILITY

Empty

E. ETHICAL STATEMENT

Empty

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